From Generating to Verifying

A reference-free measure of deck usability, used as a control signal in paper-to-slide generation

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1. Background

A conference talk gives an author ten to fifteen minutes to communicate work that occupies eight pages of a paper. The deck is therefore not a summary of the paper; it is a compression of it into an argument that an audience can follow in real time. Producing that compression is slow: the author must decide what to keep, find the figure that carries each claim, and fit both into a page that remains legible from the back of a room.

Language models can now produce a plausible deck from a paper in under a minute. The question this project starts from is not whether they can, but whether the result is usable — and whether the system itself can tell.

That second question matters because the failures are not subtle. A text box that runs past the slide edge, two elements that overlap, a caption that sits on a dark background, a page carrying two hundred and fifty words: these are mechanical failures, visible the moment the deck is projected. They are also the failures that a language model cannot see, because it never renders what it has written.

2. How this is done today

Work in this area has settled into two families, with an evaluation literature growing alongside them.

Generation. PPTAgent treats deck creation as editing rather than authoring: it analyses reference presentations, extracts slide-level schemas, and emits edit actions. SlideGen, SlideLab and ArcDeck add narrative planning and multimodal grounding. These systems produce convincing decks, and on blind preference tests they are competitive with commercial tools.

Repair. ReDeck decomposes revision into atomic edit actions and returns a renderer-derived observation after each one. PPTPilot, introduced with PPTArena, runs a plan-edit-check loop in which a vision-language model judges the result. SLIDEFORGE keeps slides as structured artefacts so that edits preserve native editability.

Evaluation. SlideAudit (UIST '25) provides an expert taxonomy of 27 slide design flaws over 2,400 annotated slides. DECKBench (KDD '26) scores decks against reference decks with similarity metrics and an LLM judge. PresentBench scores generation with fine-grained rubrics.

Table 1. What each body of work provides, and what it leaves open.

Work	Provides	Leaves open
PPTAgent, SlideGen, SlideLab	Strong generators; narrative planning	Whether the output is usable; local failures
ReDeck, PPTPilot, SLIDEFORGE	Repair loops with rendered feedback	Which repairs helped, and by how much
SlideAudit	Flaw taxonomy; 2,400 annotated slides	No loop; image-based; no per-class repair analysis
DECKBench	Deck-level scoring against a reference	Needs a target deck; ranks systems, does not drive one

3. What is good, and what is not

Three things are genuinely solved. Decks look convincing: layout is stable and styling consistent. Output is increasingly editable, because work has moved into the .pptx object layer. And evaluation has matured from holistic scores to task-based measures — answering questions using only the artefact, or gating readiness on explicit criteria.

Four things are not.

- Attribution is missing. ReDeck and PPTPilot both run repair loops, but neither reports which class of defect responded to repair and which did not. A deck-level gain of ten points tells us the loop helped; it does not tell us where.
- Measurement is post-hoc. SlideAudit and DECKBench score finished decks. A score that arrives after generation cannot steer generation.
- The instruments need references or labels. DECKBench requires a target deck; SlideAudit requires human annotation of images. Neither can be computed on an arbitrary .pptx at run time.
- Thresholds are conventions. The numbers used to decide whether a page is too dense — six bullets, thirty words, 24-point type — are widely repeated and rarely evidenced, as Section 6 sets out.

4. Our reading of the evidence

Put together, the literature describes a field that is good at producing decks and good at scoring them afterwards, but that has not closed the loop between the two. The instrument arrives too late to be used, and it is not decomposable when it does arrive.

That is the gap this project addresses. It is narrow on purpose: not another generator, not a new taxonomy, but a measurement layer that can be computed during generation and a study of what that measurement is worth.

5. Why this is worth doing

First, the failures are mechanical, so they can be decided deterministically. Whether a shape lies outside the slide, whether two text boxes intersect, whether a colour pair reaches a contrast ratio: these are arithmetic, not judgements. A rule can be audited; a model's opinion cannot.

Second, usability is a precondition. A deck that cannot be read fails regardless of how well its content was selected. Improving content on an unreadable page does not produce a better talk.

Third, the result transfers. If the measurement is reference-free and computed on the object model, it can be applied to any deck produced by any system — including the baselines this project compares against. That is what makes the comparison fair, and what makes the artefact reusable.

6. Convention, not evidence

Practitioners are not short of rules. The IEEE NMDC author guidelines, for example, state that authors should "not use font sizes smaller than 24 point" and that a text-only page should carry "no more than 30 words and no more than 6 lines". The advice is clear, widely followed, and entirely without cited evidence. No study is referenced; other conferences give different numbers.

The rules also disagree with each other. The six-by-six rule allows six words on each of six lines — thirty-six words in total — while the guideline above caps the same page at thirty. Both are stated with confidence; neither cites a study, and the practitioner literature has begun to argue that the six-by-six rule should be retired. When two widely used conventions differ by twenty per cent and neither can point to evidence, what is being transmitted is habit, not knowledge.

It is worth separating three kinds of anchor, because they carry different weight and the design must treat them differently.

Table 2. Three kinds of anchor. Only the first is evidence in the ordinary sense; the third is evidence only if the sample supports it.

Anchor	Example	Where it comes from	Can it support a measurement?
Formal standard	WCAG 2.2 AA contrast, 4.5:1	Perceptual research, normatively fixed	Yes — testable against users
Practitioner convention	24 pt; 30 words; 6 lines	Repeated across guides, no citation chain	No — a default, not a finding
Empirical value	Overlap greater than 5%	Fitted to a particular corpus	Only with sufficient data

Conventions persist for reasons unrelated to validity. They are cheap to state, easy to check, and plausible on their face. Nobody is held accountable when one is wrong, because the cost of a wrong threshold is invisible: a page that is slightly too dense still presents, and the audience simply reads less. A guideline can survive for decades on plausibility alone.

What would it take to validate one? Vary the value, hold content and audience constant, and measure comprehension or preference — a design with severe confounds, since content quality dominates any effect of density. That study does not exist for slides, and it is not the contribution

of this project. The honest position is therefore not that the conventions are wrong, but that they cannot bear the weight of a measurement instrument.

The consequence is methodological. There is no validated value to fit, and forty calibration slides cannot support six fitted thresholds — roughly six slides per gate — so a value chosen that way would be an artefact of the sample. The design separates the gates by provenance instead:

Table 3. Gate provenance. Two gates are fixed by rule or published standard; the remaining four are swept, and none is fitted.

Gate	Rule	Threshold	Provenance
G1 Containment	Every shape box inside the slide	0 pt	Binary rule — no free parameter
G4 Contrast	WCAG 2.2 AA ratio, fill against background	4.5:1 (3:1 large)	Published standard
G3 Font size	Body and title size, by slide role	role-specific	Convention; swept, not fitted
G6 Density	Characters and bullets, by slide role	role-specific	Convention; swept, not fitted
G2 Overlap	Text intersection over smaller box	> 5%	Empirical; swept
G5 Text fit	Rendered text height over box height	≤ 0.95	Renderer tolerance; swept

Swept, not fitted, means the value is reported across a range and a conclusion is only accepted if it survives the whole range. The two standard-anchored gates have no free parameter at all. Because this project generates the decks, the role of each slide — title, agenda, content, figure, conclusion — is known from the outline, so density and type limits can differ by role without a classifier. That removes the objection that a title slide and a methods slide cannot share one density limit: they no longer do.

7. Plan and schedule

The work is organised into four streams. Each produces something reportable on its own, so that a slip in one does not leave the project without a result.

- W1 — Metric. Six gates implemented on the .pptx object model; SRR and deck-ready computed for every deck.
- W2 — Repair. A decision procedure per violation class, cheapest reversible action first, with a guardrail that forbids adding content or changing numbers.
- W3 — Control. The loop that applies W2 under W1's signal: greedy, iterative, then budget-aware.
- W4 — Evidence. Detector validation against human labels; discriminative validity across systems; a controlled ablation of the verifier; a human study.

Two checks run before the main study, because a failure in either invalidates the rest. Discriminative validity asks whether SRR orders systems as it should — human-made decks above

PPTAgent above a naive pipeline. If human decks do not score highest, the gates are mis-specified and must be revised before any loop is run. Convergent validity asks whether SRR tracks human preference; that correlation is pre-registered, and it will be reported whether or not it holds.

Two departures from the brief's suggested schedule are deliberate. The brief places inspection in stage 5, weeks 12 to 14; this project begins it in week 5, because the metric has to be validated before the loop that depends on it is built. Second, the brief's stage 6 comparison names accuracy, coverage, readability, design, latency and cost. Coverage — whether the paper's key claims survive into the deck — is measured explicitly, with five to eight claims annotated per paper, so that all six dimensions are reported.

Table 4. Schedule against the University timeline.

Weeks	Stream	Deliverable	Milestone
W1–2	Scope and literature	38 papers reviewed; gap fixed	Proposal due 5 Nov 2026
W3–4	Environment and spikes	Parse and render working	Workshop 1, 9 Nov
W5–7	Rulebook and detectors	Six gates; 40-slide calibration	Discriminative validity check
W8–11	Layout and rendering	Repair actions; 110-slide test set	First SRR comparison
W12–14	Calibration and loop	Sweeps frozen; loop converged	Interim report, 7 Jan 2027
W15–18	Experiments and study	Ablation; 20-participant study	Preference correlation
W19–20	Report and demonstration	Report, demo, slides	Final, 8 Apr 2027; presentation 9–21 Apr

8. Disclosure

Generative AI tools were used for drafting and for code generation. All design decisions, the metric definitions, the evaluation protocol, the annotation and the analysis are the author's. The tools used and the extent of their use are recorded in a log, submitted with the report in line with the University's student guide on the use of GenAI.

References

Chen, H., Zhou, X., Zhou, W., et al. (2026). X+Slides: Benchmarking audience-conditioned slide generation. arXiv:2606.19256.

Hu, H., Dang, C., Liu, Y., et al. (2026). PosterMELD: Multi-agent paper-to-poster generation with editable print-ready outputs. arXiv:2608.02218.

Huang, Z., Liu, X., Hu, T., Zhang, K., & Liu, Y. (2025). PPTBench: Holistic evaluation of LLMs for PowerPoint layout and design understanding. arXiv:2512.02624.

IEEE Nanotechnology Materials and Devices Conference. (2021). Presentation preparation guidelines and requirements. https://ieeenmdc.org/past-conferences/nmdc-2021/call-for-papers/presentation-preparation-guidelines-and-requirements/

Jang, D., Heisler, M. L., Xing, L., et al. (2026). DECKBench: Benchmarking multi-agent frameworks for academic slide generation and editing. arXiv:2602.13318.

Ofengenden, M., Man, Y., Pang, Z., Gui, L.-Y., & Wang, Y.-X. (2025). PPTArena: A benchmark for PowerPoint editing. arXiv:2512.03042.

Ozden, T. C., VS, S., Horoz, F., et al. (2026). Narrative-driven paper-to-slide generation via ArcDeck. arXiv:2604.11969.

Pang, W., Lin, K. Q., Jian, X., He, X., & Torr, P. (2025). Paper2Poster: Towards multimodal poster automation from scientific papers. arXiv:2505.21497.

Tian, M., Zeng, Z., Yang, Y., et al. (2026). ReDeck: Step-level render-grounded refinement for document-to-slide generation. arXiv:2609.00194.

Vats, V., Sharma, K., & Wang, Y. (2026). SlideLab: Audience-centered scientific slide generation and evaluation. arXiv:2609.30294.

Zhang, Z. J., Chen, R., Zhong, M., & Wobbrock, J. O. (2025). SlideAudit: A dataset and taxonomy for automated evaluation of presentation slides. arXiv:2508.03630.

Zheng, H., Guan, X., Kong, H., et al. (2025). PPTAgent: Generating and evaluating presentations beyond text-to-slides. arXiv:2501.03936.

Zheng, H., Mo, G., Yan, X., et al. (2026a). DeepPresenter: Environment-grounded reflection for agentic presentation generation. arXiv:2602.22839.

Zheng, H., Wang, F., Xiong, T., et al. (2026b). SLIDEFORGE: An LLM agent for controllable editing of slides as structured artifacts. arXiv:2609.03109.